

Product Dissection for Zerodha

India's Largest Retail Stock Broker

Domain	FinTech / Stock Broking
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Product Dissection for Zerodha

Company Overview:

Zerodha, founded in 2010 by Nithin Kamath and Nikhil Kamath, is India's largest retail discount stockbroker by active client count. The name "Zerodha" is a combination of "Zero" and "Rodha" — the Sanskrit word for barrier — encapsulating the company's founding mission: to eliminate the barriers that kept ordinary Indians away from financial markets.

Headquartered in Bengaluru, Zerodha disrupted the traditional broking industry by introducing a flat fee of ₹20 per executed order, regardless of trade size, at a time when full-service brokers charged percentage-based commissions. Today, Zerodha serves over 13 million active clients, accounts for approximately 15-17% of India's daily retail trading volumes, and has become a profitable, bootstrapped company with annual revenues exceeding ₹4,700 crore — all without raising external venture capital.

Zerodha's ecosystem includes Kite (its flagship trading platform), Console (a portfolio analytics and reporting dashboard), Coin (a direct mutual fund investment platform), Varsity (a free financial education portal), and Streak (an algorithmic trading tool). This case study focuses primarily on Kite and Console, which form the core of Zerodha's product experience for active traders.

Product Dissection and Real-World Problems Solved by Zerodha:

Zerodha, India's pioneering discount stockbroker, has addressed deep-rooted real-world problems in India's retail investing and trading landscape through a series of thoughtfully designed product features. By combining technology with a user-first philosophy, Zerodha has made stock market participation accessible, transparent, and operationally efficient for millions of Indians.

At the heart of Zerodha's impact is its flagship trading platform, Kite. Before platforms like Kite existed, retail traders in India were forced to use bulky, installation-heavy desktop software offered by traditional brokers, or rely on phone-based order placement through relationship managers. These systems were slow, prone to error, and entirely inaccessible on mobile devices. Kite solved this by delivering a browser-based and mobile trading experience with real-time streaming data, TradingView-powered advanced charting with over 100 technical indicators, and a clean minimalist interface that reduced the cognitive overhead of order placement to just a few clicks.

Another major innovation is the Good Till Triggered (GTT) order system. India's stock exchanges, unlike some global counterparts, do not natively support persistent conditional orders. Traders wishing to buy or sell at a specific price had to be physically present and manually place orders each day. Zerodha's GTT feature built a conditional order management layer on top of exchange infrastructure, allowing orders to be set once and triggered automatically when price conditions are met — even days or weeks later. This transformed how positional and swing traders operated, removing the need for constant screen monitoring.

Console, Zerodha's analytics dashboard, addressed the longstanding opacity in brokerage reporting. Traditional brokers provided minimal information about charges, P&L breakdowns, and tax implications. Console provides granular, CA-ready P&L reports, capital gains statements, turnover reports for tax audit purposes, and complete brokerage charge breakdowns — giving traders unprecedented financial transparency. For a market where tax compliance around trading income has historically been complex, this feature alone represents a significant real-world problem solved.

In conclusion, Zerodha's product design has successfully tackled real-world problems by building a platform that democratizes market participation, automates operational complexity, and delivers financial transparency that was previously unavailable to India's retail trading community. Through Kite, GTT, Console, and its broader ecosystem, Zerodha has reshaped the Indian broking landscape and continues to set the benchmark for technology-first financial services.

Case Study: Real-World Problems and Zerodha's Innovative Solutions

Zerodha, India's leading discount stockbroker, has not only democratised access to financial markets but has also addressed significant real-world challenges faced by retail traders and investors through its innovative product features. The following case study draws on direct personal experience as an active F&O and intraday trader on Zerodha for over a year, using Kite and Console as primary tools.

Problem 1: The Inaccessibility of Professional Trading Tools

Real-World Challenge: When I first began trading in equity derivatives, the tools available through traditional brokers were either prohibitively expensive, required complex desktop installations, or were simply too slow for intraday use. The interface intimidation alone was enough to discourage many new participants from actively engaging with F&O markets. The cognitive overhead of learning a clunky system while simultaneously managing risk was a significant barrier.

Zerodha's Solution: Kite's interface was a revelation. Order placement for an F&O contract was reduced to three clicks. The integrated TradingView charting meant I no longer needed a separate charting platform — everything was in one tab. The real-time streaming watchlist updated tick-by-tick without page refreshes. Within a week of switching to Kite, my order placement time dropped from several minutes to under ten seconds — a difference that is not cosmetic for intraday trading but directly impacts execution quality and P&L.

Zerodha's Approach: Zerodha deliberately designed Kite around the principle of removing everything that does not serve the trade. The minimalist UI philosophy — inspired by international platforms like Robinhood but adapted for India's complex multi-segment market — solved the problem of interface intimidation and made professional-grade tools accessible to retail participants without a learning curve steep enough to deter them.

Problem 2: The Absence of Persistent Conditional Orders

Real-World Challenge: As a positional F&O trader, I frequently identify entry and exit levels in advance based on technical analysis — but I cannot monitor the screen continuously throughout the trading day. Indian exchanges do not support long-duration conditional orders natively, which meant that every day I either had to be glued to the screen watching for my price levels or risk missing trades entirely. During a particularly volatile session, I identified ₹18,200 on the Nifty index as a strong support level where I wanted to initiate a long position. I had a meeting I could not reschedule.

Zerodha's Solution: I set a GTT (Good Till Triggered) order at ₹18,200 for a Nifty futures contract the night before. By afternoon, the market dipped to that level, the GTT triggered, and my order was executed — while I was entirely away from the screen. The trade worked out profitably. Without GTT, I would have

missed the setup entirely or been forced to choose between the meeting and the trade. This is not a convenience feature — for positional traders it is operationally transformative.

Zerodha's Approach: GTT functions by having Zerodha's servers continuously monitor market prices and fire a regular exchange order the moment the trigger condition is met. This architecture effectively builds a conditional automation layer on top of exchange infrastructure that does not natively support it, solving a market structure gap without requiring traders to use third-party algorithmic trading software.

Problem 3: Opacity in Brokerage Reporting and Tax Compliance

Real-World Challenge: After my first year of active F&O trading, I faced the daunting task of computing my tax liability. F&O income is treated as business income in India, requiring proper turnover computation, expense deduction, and in some cases a tax audit. The statements provided by traditional brokers were minimal — daily contract notes with no consolidated view, no P&L by instrument, and no computed turnover figures that the Income Tax Act requires. Reconstructing this manually was an error-prone, week-long exercise.

Zerodha's Solution: Console's tax P&L report generated in seconds what would have taken days to compile manually. It provided segment-wise P&L, absolute turnover (the sum of absolute profit and loss across all F&O trades as required by the IT Act), realised vs unrealised breakdowns, and brokerage charge summaries — all downloadable in formats that my CA could directly use for ITR filing. The console also showed me the exact impact of brokerage, STT, GST, SEBI charges, and stamp duty on each trade — information that traditional brokers never surfaced.

Zerodha's Approach: Console was built with the understanding that for active traders, financial transparency is not a nice-to-have — it is a regulatory necessity. By designing reporting around actual tax compliance requirements (ITR-3, turnover computation, capital gains segregation), Zerodha solved a problem that affects every active trader in India at year-end, and positioned itself as a platform that genuinely supports its users beyond the moment of trade execution.

In conclusion, Zerodha's journey from a discount broker to a full-stack financial services platform is a testament to its ability to identify real-world problems and address them through precise product design. The case study above — grounded in personal experience — demonstrates that Kite, GTT, and Console are not features built for marketing value but solutions engineered for the specific operational and regulatory challenges faced by Indian retail traders.

Top Features of Zerodha:

- **Kite Trading Platform:** Zerodha's browser and mobile-based trading platform offering real-time streaming quotes, TradingView-integrated charting with 100+ indicators, multi-segment order placement (Equity, F&O, Commodity, Currency), and a minimalist UI designed for fast execution. Available on web, Android, and iOS at zero platform fee.
- **GTT — Good Till Triggered Orders:** A conditional order system that allows traders to set trigger-based orders that persist until the trigger price is hit or the order expires. Supports both single trigger (buy/sell at a price) and OCO (One Cancels Other) triggers for simultaneous stop-loss and target orders. Fills the gap created by the absence of native long-duration conditional orders on Indian exchanges.
- **Console — Analytics Dashboard:** A comprehensive portfolio and reporting platform providing P&L statements, capital gains reports, segment-wise turnover computation for tax purposes, brokerage charge breakdowns, and holdings analytics. Designed to meet actual tax compliance requirements for Indian traders filing under ITR-3 as business income.
- **Coin — Direct Mutual Fund Platform:** An integrated platform for investing in direct mutual funds at zero commission, bypassing distributors and regular plans. Enables SIP automation, portfolio tracking, and NAV-based order placement — all within the Zerodha ecosystem using the same login credentials.
- **Margin and Pledge System:** Allows traders to pledge equity holdings and ETFs held in their demat account as collateral for F&O margin, subject to exchange-mandated haircuts. Provides capital efficiency by enabling simultaneous long-term investing and active derivatives trading without liquidating either position.
- **Kite Connect API:** A REST API that exposes the full Kite trading functionality — order placement, market data streaming, portfolio access — to developers and algorithmic traders. Enables retail participants to build and deploy automated trading strategies without institutional infrastructure.

Schema Design Based on Top Features

The schema for Zerodha involves multiple entities that represent different aspects of the platform. These entities include Users, Orders, Instruments, Holdings, GTT Triggers, and Positions. Each entity has specific attributes that describe its properties and relationships with other entities. The schema is designed to reflect Zerodha's core business operations as a multi-segment stock broker serving active retail traders in India.

Schema Description:

User Entity:

Attribute	Data Type	Constraint	Description
UserID	VARCHAR(20)	PRIMARY KEY	Unique client code assigned at account opening
Full_Name	VARCHAR(100)	NOT NULL	Legal name of the account holder as per PAN
Email	VARCHAR(100)	UNIQUE, NOT NULL	Registered email for account communication
Mobile	VARCHAR(15)	UNIQUE, NOT NULL	Linked mobile number for OTP and alerts
PAN	CHAR(10)	UNIQUE, NOT NULL	Permanent Account Number for tax and KYC
Account_Type	ENUM	NOT NULL	Individual, HUF, or Corporate account type
KYC_Status	ENUM	NOT NULL	Verified, Pending, or Rejected KYC status
Registration_Date	DATE	NOT NULL	Date on which the trading account was opened
Segment_Access	SET	NOT NULL	Segments enabled: EQ, FO, CDS, COM

Order Entity:

Attribute	Data Type	Constraint	Description
OrderID	VARCHAR(30)	PRIMARY KEY	Unique order reference generated per trade
UserID	VARCHAR(20)	FK → User	The client account that placed the order
InstrumentID	VARCHAR(30)	FK → Instrument	The security or contract being traded

Attribute	Data Type	Constraint	Description
Order_Type	ENUM	NOT NULL	Market, Limit, Stop-Loss, or Stop-Loss Market
Transaction_Type	ENUM	NOT NULL	BUY or SELL
Quantity	INT	NOT NULL	Number of shares or lots in the order
Price	DECIMAL(10,2)	NULLABLE	Limit price; null for Market orders
Status	ENUM	NOT NULL	Open, Executed, Cancelled, or Rejected
Product_Type	ENUM	NOT NULL	CNC (delivery), MIS (intraday), NRML (F&O)
Timestamp	DATETIME	NOT NULL	Date and time of order placement
Exchange	ENUM	NOT NULL	NSE, BSE, NFO, MCX, or CDS
GTT_Flag	BOOLEAN	DEFAULT FALSE	Indicates if this order was triggered by a GTT

Instrument Entity:

Attribute	Data Type	Constraint	Description
InstrumentID	VARCHAR(30)	PRIMARY KEY	Unique trading symbol identifier
Trading_Symbol	VARCHAR(30)	NOT NULL	Exchange ticker symbol (e.g. RELIANCE, NIFTY25JANFUT)
Name	VARCHAR(100)	NOT NULL	Full name of the company or contract
Exchange	ENUM	NOT NULL	NSE, BSE, NFO, MCX, or CDS
Instrument_Type	ENUM	NOT NULL	EQ, FUT, OPT, ETF, or INDEX
Expiry_Date	DATE	NULLABLE	Expiry date for F&O contracts; null for equity
Strike_Price	DECIMAL(10,2)	NULLABLE	Strike price for options; null otherwise
Option_Type	ENUM	NULLABLE	CE (Call) or PE (Put) for options
Lot_Size	INT	NULLABLE	Contract lot size for F&O instruments
ISIN	CHAR(12)	NULLABLE	International Securities Identification Number

Holding Entity:

Attribute	Data Type	Constraint	Description
HoldingID	VARCHAR(30)	PRIMARY KEY	Unique identifier for each holding record
UserID	VARCHAR(20)	FK → User	The account that owns this holding
InstrumentID	VARCHAR(30)	FK → Instrument	The security held in the demat account
Quantity	INT	NOT NULL	Total number of shares held
Avg_Cost	DECIMAL(10,2)	NOT NULL	Average cost of acquisition (FIFO basis)
Current_Value	DECIMAL(12,2)	NOT NULL	Current market value based on LTP
PnL	DECIMAL(12,2)	NOT NULL	Unrealised profit or loss on the holding
Day_Change	DECIMAL(10,2)	NOT NULL	Change in value since previous day's close
Pledged_Qty	INT	DEFAULT 0	Quantity of shares pledged as F&O collateral

GTT_Trigger Entity:

Attribute	Data Type	Constraint	Description
GTT_ID	VARCHAR(30)	PRIMARY KEY	Unique identifier for the GTT trigger rule
UserID	VARCHAR(20)	FK → User	The client who created this GTT
InstrumentID	VARCHAR(30)	FK → Instrument	The instrument this GTT is set on
Trigger_Type	ENUM	NOT NULL	Single (one trigger) or OCO (One Cancels Other)
Trigger_Price	DECIMAL(10,2)	NOT NULL	Price at which the condition is evaluated
Limit_Price	DECIMAL(10,2)	NOT NULL	Limit price of the order fired on trigger
Quantity	INT	NOT NULL	Order quantity to be placed when triggered
Transaction_Type	ENUM	NOT NULL	BUY or SELL order type on trigger
Status	ENUM	NOT NULL	Active, Triggered, Cancelled, or Expired
Created_At	DATETIME	NOT NULL	Timestamp of GTT creation
Expires_At	DATE	NOT NULL	Date on which the GTT rule expires if not triggered

Position Entity:

Attribute	Data Type	Constraint	Description
PositionID	VARCHAR(30)	PRIMARY KEY	Unique identifier for the open position
UserID	VARCHAR(20)	FK → User	The trader holding this position
InstrumentID	VARCHAR(30)	FK → Instrument	The contract or security in the position
Product_Type	ENUM	NOT NULL	MIS (intraday) or NRML (overnight F&O)
Net_Qty	INT	NOT NULL	Net open quantity (positive=long, negative=short)
Buy_Avg	DECIMAL(10,2)	NOT NULL	Weighted average buy price
Sell_Avg	DECIMAL(10,2)	NULLABLE	Weighted average sell price if partially closed
Realised_PnL	DECIMAL(12,2)	NOT NULL	Booked profit or loss from closed legs today
Unrealised_PnL	DECIMAL(12,2)	NOT NULL	Mark-to-market P&L on the open position
Value_At_Risk	DECIMAL(12,2)	NULLABLE	Margin blocked for this position

Relationships:

Relationship	Cardinality	Foreign Key	Business Meaning
User places Orders	1 : Many	Order.UserID → User.UserID	One client account can place many orders across sessions
User owns Holdings	1 : Many	Holding.UserID → User.UserID	One account holds multiple securities in demat
User sets GTT Triggers	1 : Many	GTT_Trigger.UserID → User.UserID	One account can create multiple active GTT rules
User holds Positions	1 : Many	Position.UserID → User.UserID	One account can have multiple open intraday/F&O positions
Order targets Instrument	Many : 1	Order.InstrumentID → Instrument.InstrumentID	Many orders can be placed on the same instrument
Holding references Instrument	Many : 1	Holding.InstrumentID → Instrument.InstrumentID	Many accounts can hold the same security
GTT Trigger targets Instrument	Many : 1	GTT_Trigger.InstrumentID → Instrument.InstrumentID	Multiple GTTs can be set on the same instrument
Position references Instrument	Many : 1	Position.InstrumentID → Instrument.InstrumentID	Multiple users can hold positions in the same contract

ER Diagram

The Entity-Relationship diagram below vividly portrays the relationships and attributes of the entities within the Zerodha schema. This ER diagram serves as a visual representation, shedding light on the pivotal components of Zerodha's data model. By employing this diagram, a clearer understanding of the intricate interactions and connections that define the platform's data architecture is achieved.

User
string UserID <i>PK</i>
string Full_Name
string Email
string Mobile
string PAN
string Account_Type
string KYC_Status
date Registration_Date
set Segment_Access

Instrument
string InstrumentID <i>PK</i>
string Trading_Symbol
string Name
string Exchange
string Instrument_Type
date Expiry_Date
decimal Strike_Price
string Option_Type
int Lot_Size

Order
string OrderID <i>PK</i>
string UserID <i>FK</i>
string InstrumentID <i>FK</i>
string Order_Type
string Transaction_Type
int Quantity
decimal Price
string Status
string Product_Type
datetime Timestamp
bool GTT_Flag

Holding
string HoldingID <i>PK</i>
string UserID <i>FK</i>
string InstrumentID <i>FK</i>
int Quantity
decimal Avg_Cost
decimal Current_Value
decimal PnL
int Pledged_Qty

GTT_Trigger
string GTT_ID <i>PK</i>
string UserID <i>FK</i>
string InstrumentID <i>FK</i>
string Trigger_Type
decimal Trigger_Price
decimal Limit_Price
int Quantity
string Status

Position
string PositionID <i>PK</i>
string UserID <i>FK</i>
string InstrumentID <i>FK</i>
string Product_Type
int Net_Qty
decimal Buy_Avg
decimal Realised_PnL
decimal Unrealised_PnL

date Expires_At

Relationships are:

- Users place Orders – Each user can place multiple orders; each order belongs to one user.
- Users own Holdings – Each user can hold multiple securities; each holding belongs to one user.
- Users set GTT Triggers – Each user can create multiple GTT rules targeting different instruments.
- Users hold Positions – Each user can have multiple open intraday and F&O positions simultaneously.
- Orders target Instruments – Many orders can be placed on the same instrument across users.
- Holdings reference Instruments – Multiple users can hold the same security; instrument data is shared.
- GTT Triggers target Instruments – Multiple GTT rules can be set on the same instrument by different users.
- Positions reference Instruments – Multiple users can hold positions in the same contract or security.

Rationale Behind the Design

While creating this schema design, the following rationale guided the entity and relationship choices, ensuring alignment with Zerodha's operational and regulatory requirements as a SEBI-registered stock broker:

- **Separation of Order and Position entities:** Orders represent intent — they may be rejected, partially filled, or cancelled before becoming a position. Positions represent confirmed open exposure. Keeping these as distinct entities prevents data corruption and accurately reflects the two-phase reality of exchange-based execution where order acknowledgement and trade confirmation are separate events.
- **GTT Trigger as a standalone entity:** GTT triggers are conditional rules, not orders. A GTT may remain active for weeks without generating any exchange order. Modelling it as part of the Order entity would pollute the order table with inactive records and complicate order lifecycle tracking. As a standalone entity, GTT status can be monitored independently and the relationship to any generated order can be tracked via the GTT_Flag on the Order entity.
- **Instrument as a shared catalogue entity:** The Instrument entity is referenced by Orders, Holdings, GTT Triggers, and Positions. Centralising instrument data — including complex F&O attributes like Expiry_Date, Strike_Price, Option_Type, and Lot_Size — in one table avoids duplication and ensures consistency when exchange authorities update contract specifications (e.g., lot size revisions).
- **Pledged_Qty on the Holding entity:** SEBI regulations require that pledged shares be tracked separately from free shares, as pledged shares cannot be sold until the pledge is released. Maintaining Pledged_Qty as an attribute of the Holding entity — distinct from the total Quantity — ensures the system can always compute available-to-sell shares, preventing regulatory violations from over-selling pledged securities.

- **Segment_Access as a SET on the User entity:** SEBI mandates that equity, F&O, commodity, and currency segments require explicit client activation with separate risk disclosures. Storing this as a SET type allows the order management system to perform a fast, single-field permission check at order entry without an additional join to a permissions table — critical for minimising latency in the order placement flow.
- **Normalised design for regulatory compliance:** The schema avoids denormalisation that might create inconsistencies in tax reporting. Console's P&L computation, capital gains segregation, and turnover calculation depend on clean, normalised data across Order and Instrument entities. Any redundancy in this layer would risk incorrect tax computations — a compliance risk that Zerodha has explicitly designed against.

Conclusion

In this case study, we dissected the schema design and Entity-Relationship model underlying Zerodha's core platform. Zerodha has fundamentally transformed how retail participants engage with India's financial markets — moving from opaque, manual, full-service broking to transparent, technology-first, self-directed investing and trading.

The platform's intricate data model — spanning entities for Users, Orders, Instruments, Holdings, GTT Triggers, and Positions — forms the foundation for its seamless, multi-segment trading functionality. Each entity reflects a real operational or regulatory requirement of India's broking ecosystem, from SEBI's segment activation mandates to the exchange's order lifecycle flow to the Income Tax Act's turnover computation requirements.

The personal experiences documented in this case study — the transformative simplicity of Kite's interface for intraday F&O trading, the operational freedom provided by GTT during market hours, and the tax compliance relief offered by Console — demonstrate that Zerodha's schema design is not an academic exercise but a direct reflection of real trader needs. Understanding this schema provides deep insight into how Zerodha effectively manages the complexity of millions of daily transactions, positions, and regulatory reporting requirements, contributing to its continued dominance of India's retail broking landscape.